Inhibit System Recovery

Adversaries may delete or remove built-in data and turn off services designed to aid in the recovery of a corrupted system to prevent recovery.[1][2] This may deny access to available backups and recovery options.

Operating systems may contain features that can help fix corrupted systems, such as a backup catalog, volume shadow copies, and automatic repair features. Adversaries may disable or delete system recovery features to augment the effects of Data Destruction and Data Encrypted for Impact.[1][2] Furthermore, adversaries may disable recovery notifications, then corrupt backups.[3]

A number of native Windows utilities have been used by adversaries to disable or delete system recovery features:

- `vssadmin.exe` can be used to delete all volume shadow copies on a system - `vssadmin.exe delete shadows /all /quiet`
- Windows Management Instrumentation can be used to delete volume shadow copies - `wmic shadowcopy delete`
- `wbadmin.exe` can be used to delete the Windows Backup Catalog - `wbadmin.exe delete catalog -quiet`
- `bcdedit.exe` can be used to disable automatic Windows recovery features by modifying boot configuration data - `bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures & bcdedit /set {default} recoveryenabled no`
- `REAgentC.exe` can be used to disable Windows Recovery Environment (WinRE) repair/recovery options of an infected system
- `diskshadow.exe` can be used to delete all volume shadow copies on a system - `diskshadow delete shadows all` [4] [5]

On network devices, adversaries may leverage Disk Wipe to delete backup firmware images and reformat the file system, then System Shutdown/Reboot to reload the device. Together this activity may leave network devices completely inoperable and inhibit recovery operations.

On ESXi servers, adversaries may delete or encrypt snapshots of virtual machines to support Data Encrypted for Impact, preventing them from being leveraged as backups (e.g., via `vim-cmd vmsvc/snapshot.removeall`).[6]

Adversaries may also delete "online" backups that are connected to their network – whether via network storage media or through folders that sync to cloud services.[7] In cloud environments, adversaries may disable versioning and backup policies and delete snapshots, database backups, machine images, and prior versions of objects designed to be used in disaster recovery scenarios. [8][9]

ID: T1490

Sub-techniques: No sub-techniques

ⓘ Tactic: Impact

ⓘ Platforms: Containers, ESXi, IaaS, Linux, Network Devices, Windows, macOS

ⓘ Impact Type: Availability

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Procedure Examples

ID	Name	Description
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks, the adversaries deleted Windows Volume Shadow Copies using `vssadmin delete shadows`.[10]
S1129	Akira	Akira will delete system volume shadow copies via PowerShell commands.[11][12]
S0640	Avaddon	Avaddon deletes backups and shadow copies using native system tools.[13][14]
S0638	Babuk	Babuk has the ability to delete shadow volumes using `vssadmin.exe delete shadows /all /quiet`.[15][16]
S1136	BFG Agonizer	BFG Agonizer wipes the boot sector of infected machines to inhibit system recovery.[17]
S0570	BitPaymer	BitPaymer attempts to remove the backup shadow files from the host using `vssadmin.exe Delete Shadows /All /Quiet`.[18]
S1070	Black Basta	Black Basta can delete shadow copies using vssadmin.exe.[19][20][21][22][23][24][25][26][26][27]
G1043	BlackByte	BlackByte resized and deleted volume shadow copy files to prevent system recovery after encryption.[28][29]
S1181	BlackByte 2.0 Ransomware	BlackByte 2.0 Ransomware modifies volume shadow copies during execution in a way that destroys them on the victim machine.[30]
S1180	BlackByte Ransomware	BlackByte Ransomware deletes all volume shadow copies and restore points among other actions to inhibit system recovery following ransomware deployment.[31]
S1068	BlackCat	BlackCat can delete shadow copies using `vssadmin.exe delete shadows /all /quiet` and `wmic.exe Shadowcopy Delete`; it can also modify the boot loader using `bcdedit /set {default} recoveryenabled No`.[32]
S0611	Clop	Clop can delete the shadow volumes with `vssadmin Delete Shadows /all /quiet` and can use bcdedit to disable recovery options.[33]
S0608	Conficker	Conficker resets system restore points and deletes backup files.[34]
S0575	Conti	Conti can delete Windows Volume Shadow Copies using `vssadmin`.[35]
S1111	DarkGate	DarkGate can delete system restore points through the command `cmd.exe /c vssadmin delete shadows /for=c: /all /quiet"`.[36]
S0673	DarkWatchman	DarkWatchman can delete shadow volumes using `vssadmin.exe`.[37]
S0616	DEATHRANSOM	DEATHRANSOM can delete volume shadow copies on compromised hosts.[38]
S0659	Diavol	Diavol can delete shadow copies using the `IVssBackupComponents` COM object to call the `DeleteSnapshots` method.[39]
S0605	EKANS	EKANS removes backups of Volume Shadow Copies to disable any restoration capabilities.[40][41]
S1247	Embargo	Embargo has cleared files from the recycle bin by invoking `SHEmptyRecycleBinW()` and disabled Windows recovery through `C:\Windows\System32\cmd.exe /q /c bcdedit /set {default} recoveryenabled no`.[42]
S0618	FIVEHANDS	FIVEHANDS has the ability to delete volume shadow copies on compromised hosts.[38][43]

ID	Name	Description
S0260	InvisiMole	InvisiMole can can remove all system restore points.[49]
S0389	JCry	JCry has been observed deleting shadow copies to ensure that data cannot be restored easily.[50]
S1199	LockBit 2.0	LockBit 2.0 has the ability to delete volume shadow copies on targeted hosts.[51][52]
S1202	LockBit 3.0	LockBit 3.0 can delete volume shadow copies.[53][54][55]
S0449	Maze	Maze has attempted to delete the shadow volumes of infected machines, once before and once after the encryption process.[56][57]
G1051	Medusa Group	Medusa Group has deleted recovery files such as shadow copies using `vssadmin.exe`.[58][59][60][61]
S1244	Medusa Ransomware	Medusa Ransomware has deleted recovery files such as shadow copies using `vssadmin.exe`.[58][59][60][61]
S0576	MegaCortex	MegaCortex has deleted volume shadow copies using `vssadmin.exe`.[62]
S0688	Meteor	Meteor can use `bcdedit` to delete different boot identifiers on a compromised host; it can also use `vssadmin.exe delete shadows /all /quiet` and `C:\\Windows\\system32\\wbem\\wmic.exe shadowcopy delete`.[63]
S1135	MultiLayer Wiper	MultiLayer Wiper wipes the boot sector of infected systems to inhibit system recovery.[17]
S0457	Netwalker	Netwalker can delete the infected system's Shadow Volumes to prevent recovery.[64][65]
S0365	Olympic Destroyer	Olympic Destroyer uses the native Windows utilities `vssadmin`, `wbadmin`, and `bcdedit` to delete and disable operating system recovery features such as the Windows backup catalog and Windows Automatic Repair.[1]
S1162	Playcrypt	Playcrypt can use AlphaVSS to delete shadow copies.[66]
S1058	Prestige	Prestige can delete the backup catalog from the target system using: `c:\Windows\System32\wbadmin.exe delete catalog -quiet` and can also delete volume shadow copies using: `\Windows\System32\vssadmin.exe delete shadows /all /quiet`.[67]
S0654	ProLock	ProLock can use vssadmin.exe to remove volume shadow copies.[68]
S0583	Pysa	Pysa has the functionality to delete shadow copies.[69]
S1242	Qilin	Qilin can execute `vssadmin.exe delete shadows /all /quiet` to remove volume shadow copies and can disable High Availability (HA) and Distributed Resource Scheduler (DRS) in vCenter clusters.[70][71][72][73]
S0481	Ragnar Locker	Ragnar Locker can delete volume shadow copies using `vssadmin delete shadows /all /quiet`.[74]
S1212	RansomHub	RansomHub has used `vssadmin.exe` to delete volume shadow copies.[75][76]
S0496	REvil	REvil can use vssadmin to delete volume shadow copies and bcdedit to disable recovery features.[77][78][79][80][81][82][83][84][85]
S1150	ROADSWEEP	ROADSWEEP has the ability to disable `SystemRestore` and Volume Shadow Copies.[86][87]
S0400	RobbinHood	RobbinHood deletes shadow copies to ensure that all the data cannot be restored easily.[88]
S1073	Royal	Royal can delete shadow copy backups with vssadmin.exe using the command `delete shadows /all /quiet`.[89][90][91]

ID	Name	Description
G1015	Scattered Spider	Scattered Spider has stopped the Volume Shadow Copy service on compromised hosts.[93]
G1053	Storm-0501	Storm-0501 has deleted snapshots, restore points, storage accounts, and backup services to prevent remediation and restoration.[94] Storm-0501 has also impacted Azure resources through the targeting of `Microsoft.Compute/snapshots/delete`, `Microsoft.Compute/restorePointCollections/delete`, `Microsoft.Storage/storageAccounts/delete`, and `Microsoft.RecoveryServices/Vaults/backupFabrics/protectionContainers/delete`.[94]
G1055	VOID MANTICORE	VOID MANTICORE has deleted virtual machines directly from the virtualization platform.[95]
S0366	WannaCry	WannaCry uses `vssadmin`, `wbadmin`, `bcdedit`, and `wmic` to delete and disable operating system recovery features.[96][2][97]
S0612	WastedLocker	WastedLocker can delete shadow volumes.[98][99][100]
G0102	Wizard Spider	Wizard Spider has used WMIC and vssadmin to manually delete volume shadow copies. Wizard Spider has also used Conti ransomware to delete volume shadow copies automatically with the use of vssadmin.[101]

Mitigations

ID	Mitigation	Description
M1053	Data Backup	Consider implementing IT disaster recovery plans that contain procedures for taking regular data backups that can be used to restore organizational data.[102] Ensure backups are stored off system and is protected from common methods adversaries may use to gain access and destroy the backups to prevent recovery. In cloud environments, enable versioning on storage objects where possible, and copy backups to other accounts or regions to isolate them from the original copies.[103] On ESXi servers, ensure that disk images and snapshots of virtual machines are regularly taken, with copies stored off system.[104]
M1038	Execution Prevention	Consider using application control configured to block execution of utilities such as `diskshadow.exe` that may not be required for a given system or network to prevent potential misuse by adversaries.
M1028	Operating System Configuration	Consider technical controls to prevent the disabling of services or deletion of files involved in system recovery. Additionally, ensure that WinRE is enabled using the following command: `reagentc /enable`.[105]
M1018	User Account Management	Limit the user accounts that have access to backups to only those required. In AWS environments, consider using Service Control Policies to restrict API calls to delete backups, snapshots, and images.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0329	Behavioral Detection for T1490 - Inhibit System Recovery	AN0933	Process chains that use native utilities (vssadmin, wbadmin, diskshadow, bcdedit, REAgentC, wmic) with arguments to delete shadow copies, disable recovery, or remove backup catalogs
		AN0934	Shell utilities or scripts deleting `/etc/systemd/system/rescue.target`, `/etc/fstab` backups, or `/boot/efi` partitions; chattr used to block snapshot auto-recovery
		AN0935	ESXi shell or vim-cmd execution that deletes all VM snapshots using vmsvc/snapshot.removeall or rm on snapshot paths

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